Impact of a Cockroach in the Rubber Hand Illusion

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Project Background

The Rubber Hand Illusion

- Widely recognized throughout psychological studies and literature
- Has been used to alter implicit racial biases in individuals (Maister et al., 2013)

Q: What is the RHI capable of affecting?

Our Study

Q: Can a cockroach introduced to the RHI impact attitudes towards insects or animals?

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Methods

Measurements

- RHI Survey (Botvinick & Cohen 1998)
  - Introduced New Questions
- BAT (Behavioral Approach Task)
- Comfort Scale
- AAS (Animal Attitudes Scale; Herzog, 1991)
- Heart Rate

Sample RHI Questions

I felt as if the rubber hand were my hand. (--- is strongly disagree, +++ is strongly agree)

---	--	-	o	+	++	+++
○	○	○	○	○	○	○

- It seemed as though the weight or pressure I felt was caused by the cockroach touching the rubber hand.

- When the cockroach was on the rubber hand, I felt afraid or worried.

- When the cockroach was on the rubber hand, I wanted to take it off.

RHI Set-Up

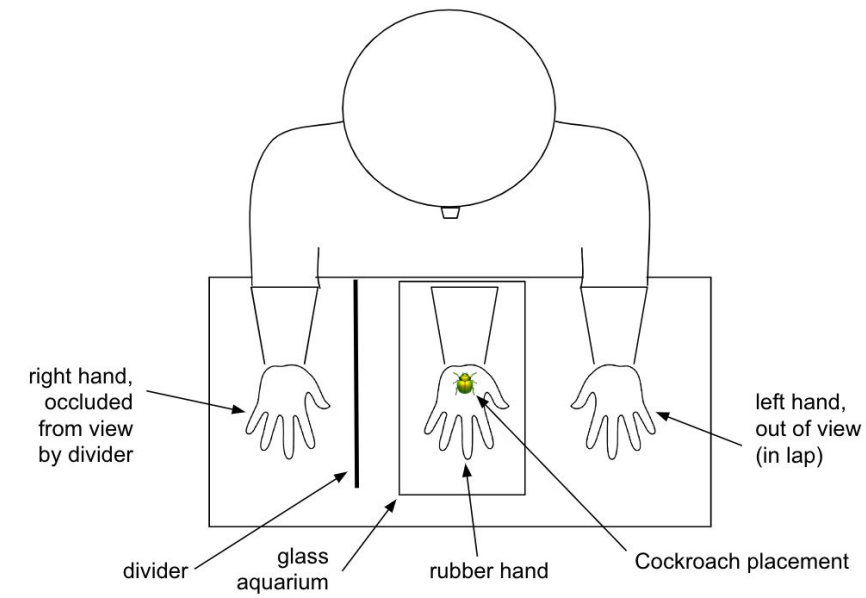
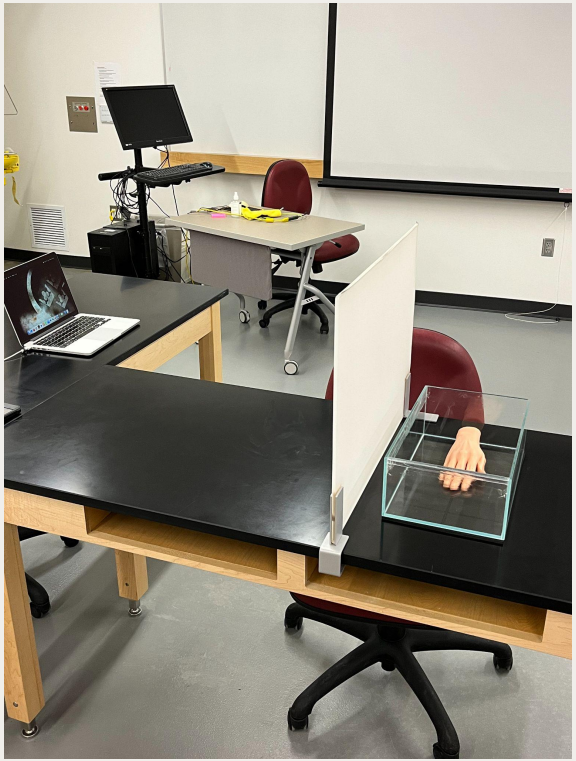
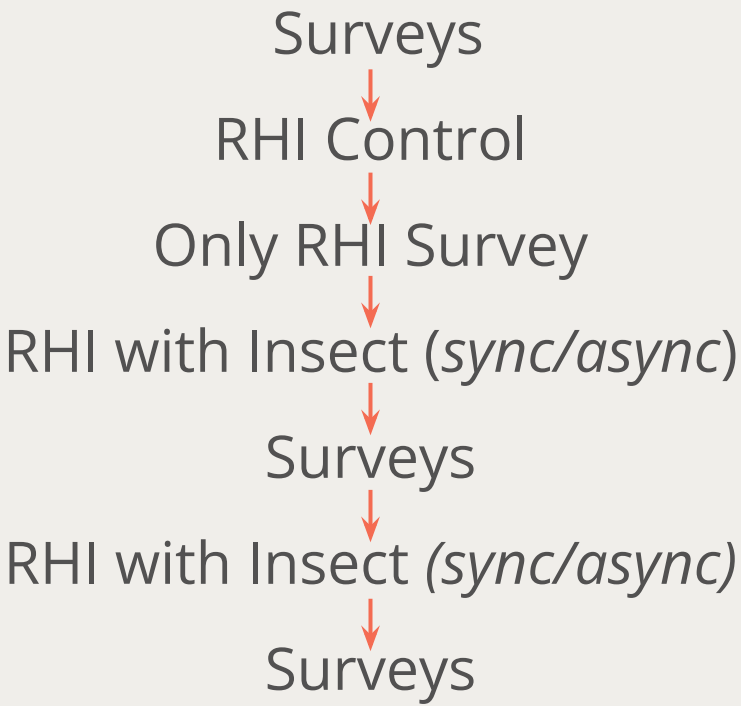


Fig. 1. Apparatus & Procedure for inducing the rubber hand illusion.
Image credit: Professor Lauren Hecht



Procedure


Surveys
RHI Control
Only RHI Survey
RHI with Insect (*sync/async*)
Surveys
RHI with Insect *(sync/async)*
Surveys

03

Results & Discussion

69.5 →80.33

Behavioral Approach Task

83 bpm →94 bpm

Participant #1673 HR

95.26→91.59

Animal Attitudes Scale

Participant #1673

“...when the paint brush tapped the ends of my fingers...felt like my hand and the rubber hand were the same”

“...I did feel like there was a weight on my hand”

“...if the paint brushes weren't totally in synch it felt like i was going numb”

Initial BAT: 27

End BAT: 76

What The Participants Had To Say

→ “At one point I thought that ‘my’ nails needed to be trimmed soon”

→ “Itchy”

→ “This time was not as much of an illusion for me. But it did distress me the most”

→ “Did not feel like the cockroach was gripping me but felt the antennae when it would touch the rubber hand”

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Looking Forward

What Now?

- Applications to exposure therapy
- Broaden use of the RHI
  - Introduced new measures
  - Introduced new methods

Thank you!

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Questions?